

SENTINEL

Everything you need to know, in plain words

Read this once. Then have someone quiz you on page 3.
It takes about fifteen minutes.

Judges pick the quietest person on the team and ask them
a basic question. If nobody can answer, the whole pitch
loses credibility no matter how good the numbers were.

PART 1

What this project actually is

A factory makes 350 computer chips in one batch. Before those chips go into a satellite, they get **baked at 125 °C for a week** — on purpose. The idea is to stress them hard so the weak ones break in the lab instead of in space. During that week, four electrical properties are measured four times: at hour 0, 24, 96 and 168.

Today the accept-or-reject rule is one line: **is the number under the limit printed on the datasheet?**
Under 50 → ship it. Over 50 → bin it.

Here is the problem, as an analogy

Imagine a doctor who only ever asks: *“is your temperature under 38 °C?”*

Your temperature went 36.5 → 36.9 → 37.4 → 37.8 across the week. You pass every single check. But you are clearly getting sick — and everyone else in your family stayed flat at 36.5 the whole time.

That chip ships. Fourteen months later the satellite dies, and nobody can go up and fix it.

So SENTINEL asks two better questions

- 1 **Is this chip behaving like its 349 siblings from the same batch?** Not “is it legal” — is it normal for its own batch.
- 2 **Where will this chip be at the end of the week, based only on its first day?** If the answer is bad, pull it early.

The one number the whole pitch rests on

All 174 defective chips in our dataset pass every datasheet check. The method the industry uses today catches **zero** of them.

Ours catches **81%**, while wrongly rejecting only **6%** of good chips.

PART 2

What we built

Three main pieces

Module A the detector	Works out what normal looks like for that specific batch , then flags chips drifting away from their own neighbours. Not against a fixed rulebook — against the siblings.
Module B the fortune teller	Using only day 0 and day 1, predicts where a chip will be on day 7. If it looks bad you pull that chip after 24 hours instead of 168. Ovens are expensive and always full, so that saves 144 oven-hours per chip pulled — about 15,000 hours across our run.
The explanation layer worth a third of the marks	Every rejection comes with a plain-English reason, not a probability. Plus a one-page PDF per rejected chip — the thing a quality inspector signs their name on.

And around all that

A live web console you can click through, a working API a factory could actually call, 150 automated tests, and a one-command demo that regenerates every number from scratch.

An example of what a rejection looks like

Reason code R-101

“Iddq at 168h is 9.3 robust sigma above the lot median (53.07 μ A vs lot median 10.8 μ A). This also EXCEEDS the datasheet limit of 50 μ A — a hard reject on static limits alone.”

Notice it gives **the chip's real number and the batch's normal number**, in real units. That is what makes it something a human can check and sign.

PART 3

The ten words you must know

Learn these ten and you are safe. Say them out loud to a teammate — do not just read them. This is the single most common way hackathon teams lose points.

TERM	WHAT YOU SAY IF ASKED
Lot	One production batch. Ours are 350 chips each, made together from the same material.
Burn-in	Baking the chips at 125 °C for a week while powered on, deliberately, so weak ones fail in the lab instead of in space.
Iddq	The current a chip draws when it is sitting idle doing nothing. Should be almost zero. It is the most sensitive sign of internal damage in a chip.
Drift	How much a measurement moves over the week. Healthy chips settle down; damaged ones keep climbing.
Latent defect	A chip that passes every test but has a flaw quietly growing inside it. This is the thing we are hunting.
Escape	A bad chip we miss and ship. In a satellite that is catastrophic — it is the number we care most about.
Overkill	A good chip we reject by mistake. Expensive, but survivable. We trade overkill to buy fewer escapes.
Datasheet limit also “USL”	The maximum value printed on the chip's spec sheet. The current industry rule is just “under the limit = ship it”.
Recall	Out of all the truly bad chips, what fraction did we catch. Ours is 81%. The existing method is 0%.
L04	Just the name of batch number 4 in our data. We deliberately made it a bad batch — 3× the defects — to check our system would notice. It did.

Two more if you are presenting the technical slides

PDA	A factory rule: if more than 5% of a batch gets rejected, the entire batch goes for review. So being trigger-happy does not just waste chips, it can scrap a good batch. That is why our system has three outcomes instead of two: ACCEPT, WATCH (ships but flagged), REJECT (only these count toward the 5%).
Wafer map	Chips are cut from a big silicon disc called a wafer. The map shows where each chip physically sat on that disc. Real defects often cluster in one spot — a scratch, a bad edge. In our data they do not cluster, and we say so. Do not claim they do.

You can ignore these
Part detail · Lot overview · Decision policy · Part explorer — these are just the menu names on our website. They are navigation labels, like “Settings”. Nobody will ask about them.

PART 4

The questions most likely to catch you out

Practise saying these answers. They are short on purpose.

“Why do you take the logarithm of the currents?”

“Currents vary enormously — one chip can leak a hundred times more than another and both be fine. On a raw scale that huge spread makes the normal range look enormous, and real outliers hide inside it. Taking the log squashes it into something sensible. Timing and voltage do not need it.”

“Why median and MAD instead of average and standard deviation?”

“Because we are hunting outliers, and outliers corrupt an average. One extreme chip drags the average toward itself and widens the spread, so it ends up hiding inside its own widened limits. The median barely moves — you would need half the batch to be defective before it shifted.”

“What is your accuracy?”

“We deliberately do not report it. About 8% of chips are defective, so a system that just says “everything is fine” scores 92% and catches nothing. We report recall instead — the fraction of bad chips actually caught.”

“Why should we trust these numbers?”

“Because we generated the data ourselves, so we know exactly which chips are truly defective and can prove our catch rate. And because we found four places our own approach did not work, and put them on a slide instead of hiding them.”

“Is this actually AI, or is it just statistics?”

“It is both, and the split is deliberate. The detector is robust statistics — median and MAD, no learning. The forecaster is a real trained machine-learning model, a gradient-boosted regressor, validated so that no chip is ever predicted by a model that saw its own batch. We tried the fancier machine-learning detector too and it scored 0.4038 against our simple statistical one at 0.4083. The simple thing won, so we shipped the simple thing and kept the other as a comparison row.”

“Why no neural network / deep learning?”

“Two thousand rows of table data. On that scale robust statistics and gradient boosting beat deep learning, and they are explainable — which is a third of the marking here. We would only justify a neural net with a number showing it beat the baseline, and it does not.”

“Your data is fake though, isn't it?”

“It is simulated, and we say so on slide one. That is a strength here: because we made it, we know the true answer for every chip, so we can actually prove our catch rate. With real factory data nobody knows which chips were secretly bad, so nobody can prove anything.”

PART 5

Every number you might be asked

If it is not on this page, do not quote it from memory.

The dataset

Chips screened	2,100
Batches (lots)	6
Truly defective chips hidden in it	174
Of those, how many pass every datasheet check	174 (100%)
Chips the old method flags — all obvious failures	55

How well we do

Recall of the existing static-limit method	0.000
Our recall (REJECT + WATCH)	0.810
Good chips wrongly rejected at that setting	6.0%
Our recall at a stricter 5% budget	0.776
Chips marked ACCEPT / WATCH / REJECT	1,785 / 210 / 105

The bad batch, and the forecast

L04 reject rate — the batch we seeded with defects	9.7%
L03 reject rate — a normal batch, for comparison	2.3%
Forecast beats simple extrapolation by	43 – 67%
Oven hours freed by rejecting at hour 24	14,976
Reason codes written	682 across 285 chips
Signed one-page PDF reports produced	105
Automated tests passing	150

PART 6

The story you tell, in order

Full slide-by-slide detail is in **SENTINEL_Slide_Blueprint.pdf** — one page per slide, with the exact numbers and the line to say. This is the shape of it.

1. The pain	A satellite dies 14 months into a 7-year mission. One bad part. No repairs possible in orbit.
2. The killer number	174 out of 174 defective chips pass every existing test. Put it huge on one slide with nothing else on it.
3. Show why	The drift chart — one chip climbing away from its batch while never crossing the legal limit. Say: <i>“static limits ask whether a part is legal; they never ask whether it is behaving.”</i>
4. What we built	Module A, Module B, and the explanations. Show a real reason code and the signed PDF.
5. The results	81% caught at 6% wrongly rejected. And the nice moment: batch L04 was secretly seeded with 3× the defects, and our system flagged it without ever being told it existed .
6. The slide that wins it	The four places our own approach did not work — see the next page.
7. The honest limit	Where we are weakest is not where our model is bad, it is where the test equipment cannot measure precisely enough . We proved it: improve the tester and the same unchanged code jumps from 65% to 92%. So the recommendation is “buy a better tester”, not “buy a better model”.

PART 7

The slide that wins it

We found four places where our own approach did not work, and we put them on a slide anyway. **No other team will have this slide.**

A technique that found nothing	The design document recommended a clever method for spotting chips whose <i>combination</i> of measurements is impossible. We measured whether that pattern exists in our data. It does not. So we dropped the method and said so.
Our 92% score, refused	Our accuracy would have been 92%, which sounds great and means nothing — guessing “all good” also scores 92%. Our code physically refuses to print it.
A bug in our own maths	One measurement looked like it improved our score by 0.09. We found it was an artefact of how we were rounding, and withdrew the improvement.
A warning that cried wolf	One of our alert codes was firing on a quarter of every batch, which is useless to an inspector. We recalibrated it down to 5%.

Say this when you show it

“Each of these made one of our numbers worse. They are on the slide because a judge who checks will find them — and because they are why you can trust the numbers that are left.”

Four things NOT to say

Never quote accuracy	Guessing “all good” scores 92%. If asked why it is missing, that sentence <i>is</i> the answer — and it earns credibility for free.
Never say “correlation break”	We checked, and that pattern does not exist in our data. A judge who checks will find that out. Say “it combines evidence across measurements” instead.
Do not oversell Module B	At hour 24 it catches roughly a third of bad chips, versus about 87% at hour 168. It buys you 144 oven-hours per chip pulled. That is a good trade on its own — overclaiming it is the first thing an expert will probe.
Do not hide that the data is simulated	Say it on slide 1. It is a strength: it is the only reason we can prove a catch rate at all.

PART 8

How anyone actually uses it

A judge 30 seconds, no install	Open the console link. Click the red batch (L04). Click the top chip in the list. They see the chart, the reasons, everything.
An engineer one command	<code>bash run_demo.sh</code> — generates the data, runs all 150 tests, prints every number in the deck, and produces the PDF reports. Identical result on any machine, because the random seed is fixed.
A real factory the actual product	Upload a spreadsheet of a fresh batch to the API and get back a verdict for every chip, with reasons. The website is just one run through that, made pretty.

Where everything lives

Code	github.com/DarkPlayz0999/Sentinel
Live console	The artifact link in your chat — share it before the pitch so judges can open it on their own screens.
Slide blueprint	docs/SENTINEL_Slide_Blueprint.pdf — one page per slide
This briefing	docs/SENTINEL_Team_Briefing.pdf

Before you rehearse

Each person explains **all ten words on page 3** out loud to another teammate. Not reading — saying. Fifteen minutes.

Then run through the deck three times, timed. Record a backup video of the demo — live demos fail, and a team that says “the wifi dropped, here is the recording” keeps its score.